

Alejandro García López

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Spain (Open to Remote & Relocation)

EXPERIENCE

Firescale Studios

Castellón, Spain

Software Engineer – Engine & Systems (C#)

Dec. 2024 – May 2026

- Architected modular, decoupled core systems using C# and dependency injection (Zenject), managing dynamic asset streams asynchronously via Addressables.
- Developed runtime cheat and diagnostic tooling to inspect internal state variables, inject parameter overrides on the fly, and log critical edge cases, reducing QA testing time by ~90%.
- Built custom Editor tooling with automated validation pipelines, standardizing cross-department data workflows.
- Profiled CPU bottlenecks, memory allocations, and garbage collection overhead using Unity Profiler and JetBrains dotTrace to ensure deterministic frame-time performance.

Catness Game Studios

Hybrid, Spain

Software Engineer – Unreal Engine (C++)

Jan. 2023 – Nov. 2024

- Engineered authority-driven state synchronization, replicated entity components, and deterministic gameplay logic in Unreal Engine 5.
- Conducted distributed stress testing and benchmarking on dedicated headless servers (~25 concurrent synchronized entities), optimizing network tick stability, RPC overhead, and bandwidth consumption.
- Profiled and diagnosed state desynchronizations, edge-case race conditions, and network bottlenecks utilizing Unreal Insights and telemetry tooling.
- Resolved 30+ TRC/XR compliance issues during console porting (PS4/PS5, Xbox Series) and optimized low-level memory budgets and multi-platform hardware performance targets for commercial releases.

PROJECTS

Autonomous Urban Traffic Simulation | Unreal Engine, C++ | [Video](#)

Jan. 2023 – June 2023

- Architected an autonomous vehicle agent framework using Finite State Machines (FSM) and spatial sensor queries to simulate dynamic traffic flow, intersection priority rules, and reactive obstacle avoidance.
- Optimized perception updates and spatial raycasting loops, achieving smooth multi-agent throughput without stalling the main simulation thread.

Real-Time Volumetric & SDF Spatial Computing | Unity, C#, HLSL | [Playable Demo](#) | [Video](#)

May 2025 – Nov. 2025

- Engineered GPU-accelerated spatial rendering and geometry querying using custom HLSL raymarching and analytical Signed Distance Functions (SDFs).
- Optimized sphere tracing algorithms and GPU compute passes to evaluate complex 3D environments at interactive frame rates.

TECHNICAL SKILLS

Core Engines & Frameworks: Unreal Engine 5, Unity, Multi-Agent Systems, Headless Execution, Physics & Spatial Partitioning

Languages: C++, C#, Python, HLSL

Simulation & Distributed Systems: High-Frequency State Synchronization, Client-Server Architecture, Deterministic Systems, RPCs, Serialization, Telemetry

3D Math & Spatial Computing: Linear Algebra, Signed Distance Functions (SDF), Raymarching, Collision Detection, Spatial Indexing (Grid/Octree)

Tooling & Profiling: Unreal Insights, JetBrains dotTrace, Unity Profiler, Git, Perforce, Linux/Windows, Rider, Visual Studio, Jira

EDUCATION

Universitat Jaume I

Castellón, Spain

Bachelor of Science in Design and Development of Video Games

Sept. 2019 – June 2023